

A STEP BY STEP GUIDE TO

Sound System Calibration

All without breaking your bank account

BONE PAGLIGARAN

Contents:

- Introduction
- The nature of sound
- How to create a calibration template
- Hardware and software requirements
- Loudspeaker positioning
- Proper loudness setting
- Loudspeaker time alignment
- Spectral balance

So what is Sound System Calibration and why do we even need to do it?

Sound System Calibration in my mind is the process of optimizing your existing sound system to the best of its capacity. It involves **(1) minimizing phase and timing issues** by properly positioning speakers and applying time delay compensation, **(2)** Applying equalization to **compensate for the room's acoustic anomalies** that affect the frequency response of the system, and **(3) set proper or ideal sound pressure levels**. Often operators or mixers calibrate simply by relying solely on their subjective hearing based on taste and experience perpetuating a culture, practice, and style that differs from mixer to mixer which 90% at the time only works for their own mixing approach. The fact of the matter is that Mixing is an entirely different discipline from System Calibration. Mixing is basically towards making a program sound close to a specific artistic direction while Calibration is more geared towards objective engineering of which the goal is to optimize a system for better performance. When we say sound engineering there has to be a balance or a marriage between both disciplines. To put it in perspective, the goal of system calibration is to achieve an optimized neutral starting point so that the mix engineer can focus on enhancing an audio program rather than trying to make things sound right based on his golden ears or taste (slight sarcasm there sorry!).

If you are able to grasp and practice this method it will guarantee a starting point that will be on a “universally ideal quality”. You will find yourself eliminating wild surgical EQ settings during soundcheck or mixing just to make things sound acceptable. No more “check mic 1-2-1-2” or manually ringing out frequencies. You will be EQ’ing to taste enhance and not to correct. Proper calibration is like proper cooking, it requires a recipe and the ingredients are properly measured and quantified, while Mixing is like where you add salt, pepper, or chili to taste when you are about to consume the dish. Suffice to say that if you foul up the recipe no amount salt and pepper tweaking can make the dish taste superb. Likewise in mixing, a properly calibrated system ensures a really good starting point for mixing.

Disclaimer!

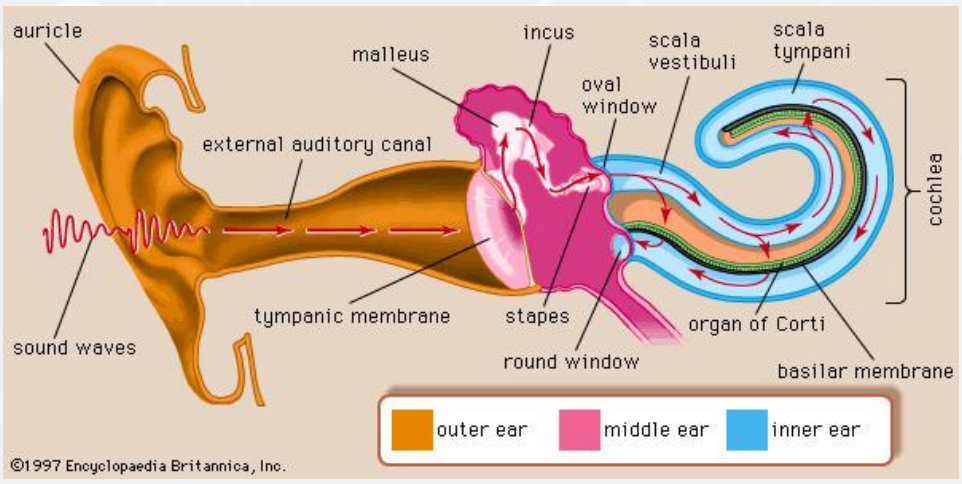
This lesson assumes that you have a decently designed system. System Design is another topic altogether. But you will be amazed at the improvement this process will bring to any existing system you optimize using this method.

The Nature of Sound

What is Sound?

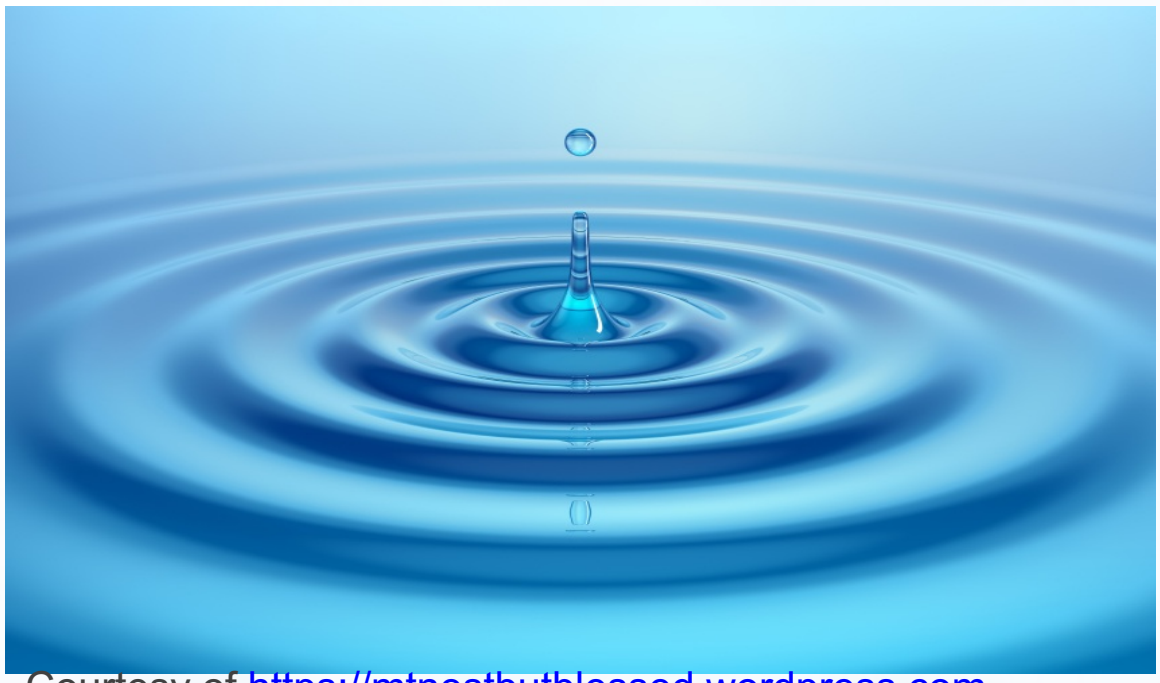
Sound is a form of energy known as acoustical energy resulting from a vibrating object. The vibrations created by an object trigger a chain reaction within the surrounding material particles thereby creating a wave motion. Such motion with enough energy enters our ear and perceives it as sound information. Sound waves enter the outer ear and travel through the external auditory canal until they reach the tympanic membrane, causing the membrane and the attached chain of auditory ossicles to vibrate. The motion of the stapes against the oval window sets up waves in the fluids of the cochlea, causing the basilar membrane to vibrate. This stimulates the sensory cells of the organ of corti, atop the basilar membrane, to send nerve impulses to the brain.

Figure 1: Hearing Mechanism



Reference: https://socratic.org/

Figure 2a: Omi-Directional Wave Pattern - Water



Courtesy of https://mtnestbutblessed.wordpress.com

Figure 2b: Omi-Directional Wave Pattern – Air



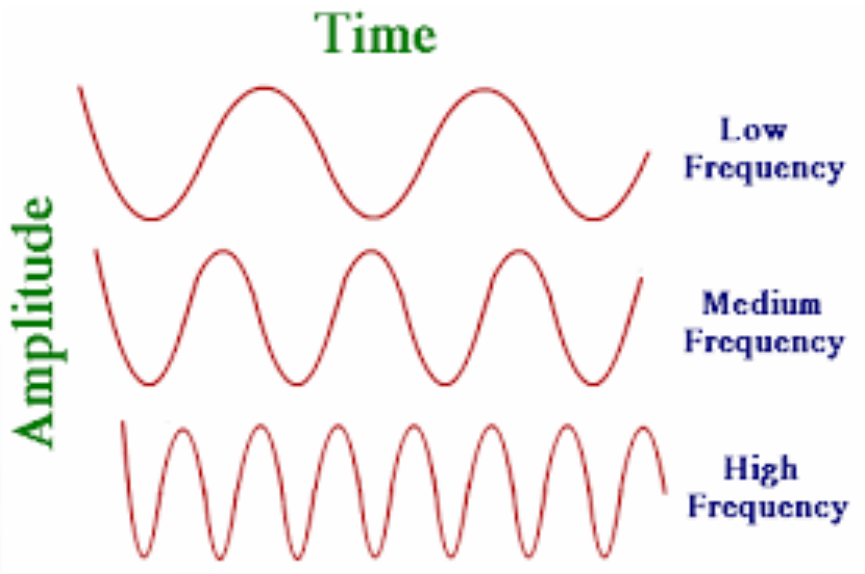
Courtesy of http://www.infobit.co

Sound Wave Characteristics

Pitch

The frequency of a sound wave is what your ear understands as pitch. A higher frequency sound has a higher pitch, and a lower frequency sound has a lower pitch.

Figure 3: Frequency



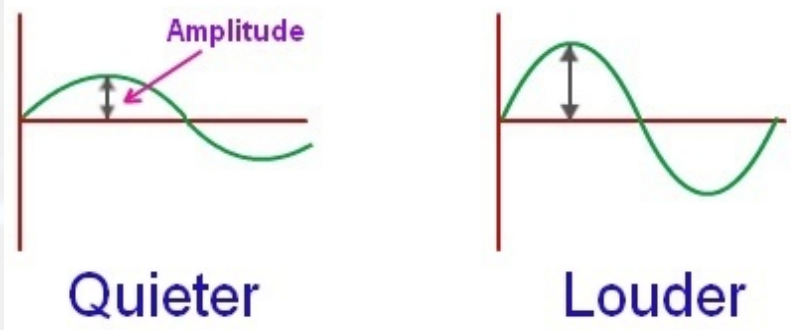
Courtesy of https://physics.stackexchange.com/

The human ear can detect a wide range of frequencies. 20 to 20,000 Hz are the audible frequencies to the human ear.

Loudness

The amplitude of a sound wave determines its loudness or volume. A larger amplitude means a louder sound, and a smaller amplitude means a softer sound. Induced corresponds to larger amplitude.

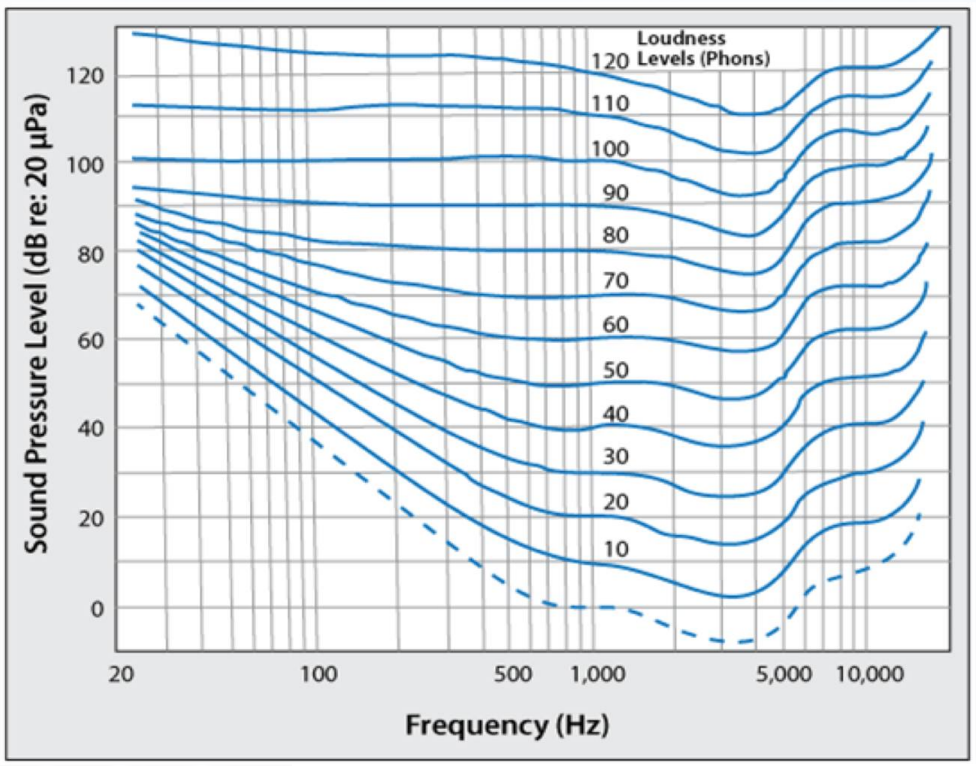
Figure 4: Amplitude



Courtesy of https://www.quora.com

The loudness of a sound is also determined by the sensitivity of the ear. The human ear is more sensitive to some frequencies than to others. The volume we receive depends on both the amplitude of a sound wave and whether its frequency lies in a region where the ear is more or less sensitive. This was determined by Fletcher-Munzon study in 1934 called the equal loudness contour.

Figure 5: Equal Loudness contour

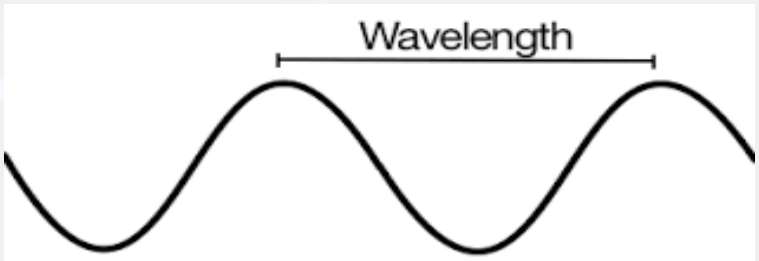


Courtesy of https://www.osha.gov

Wavelength

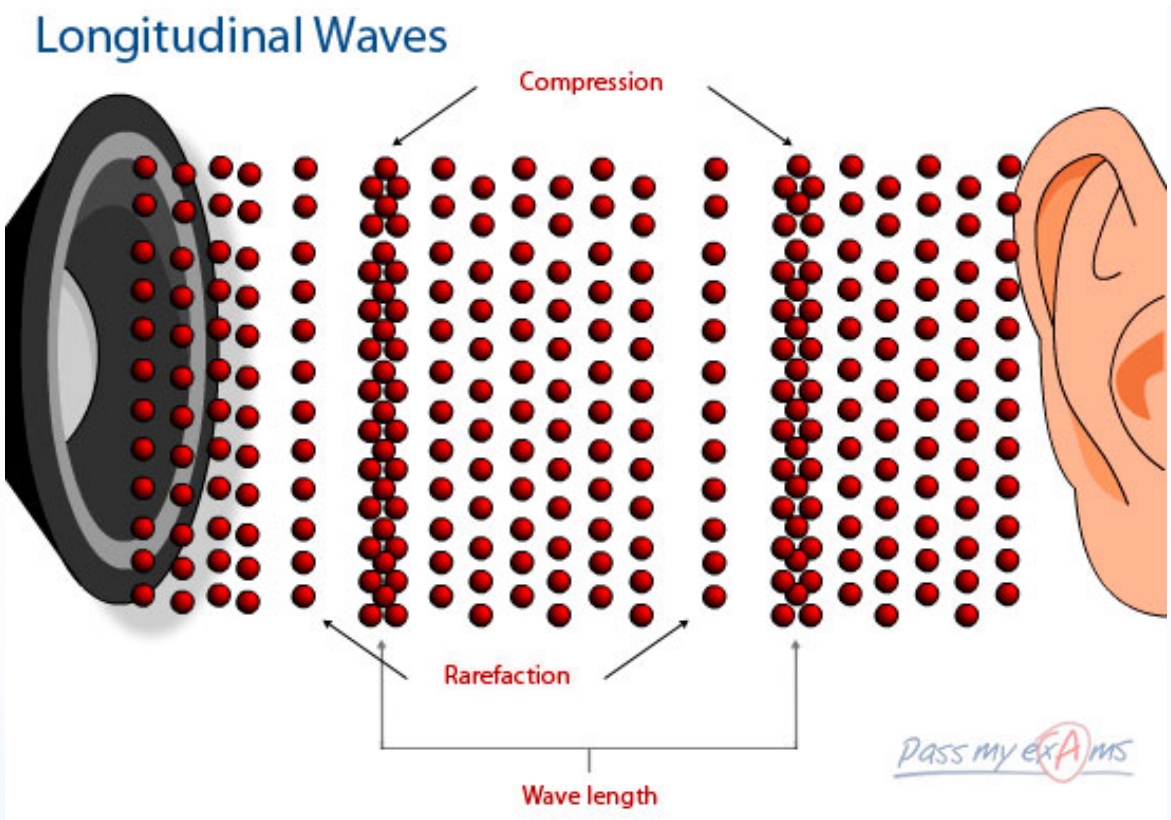
Wavelength is the distance between successive compressions or rarefactions in the air molecules.

Figure 6a: Wavelength



Courtesy of http://acousticstoday.org

Figure 6b: Wavelength



Courtesy of http://www.passmyexams.co.uk

Formula: Wavelength

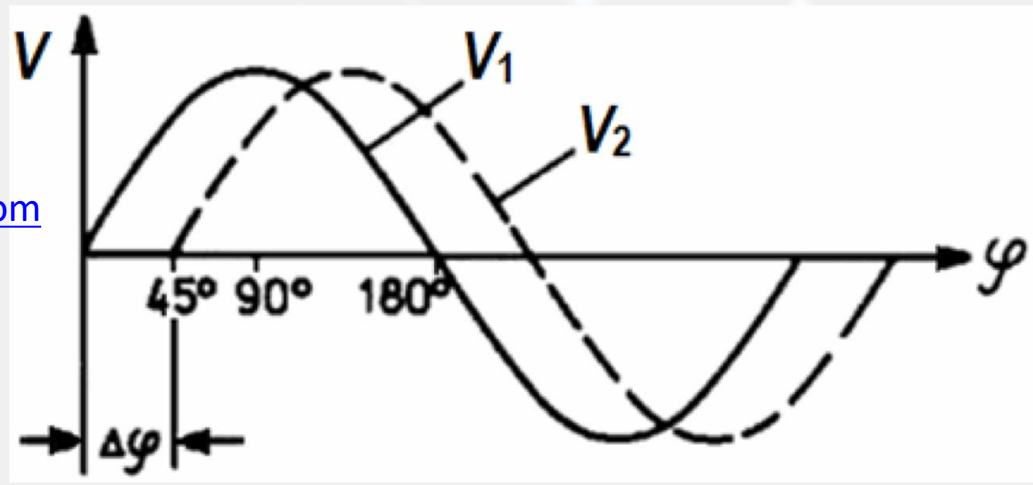
Wavelength = (340 m/s) / (19,000 Hz)
= 0.0178 m

Phase

Phase is the position of a point in time (an instant) on a waveform cycle. It is the initial angle of a sinusoidal function at its origin and is sometimes called phase offset or phase difference.

Figure 7: Phase

Courtesy of http://www.sengpielaudio.com



Velocity

The speed of sound in the air is dependent on air density, humidity, and temperature. At normal atmospheric pressure the speed of sound equation is listed below.

Formula: Speed of Sound (air)

Speed = 331 m/s + (0.6 m/s/C)•T

= 331 m/s + (0.6 m/s/C)•(20 C)

= 331 m/s + 12 m/s

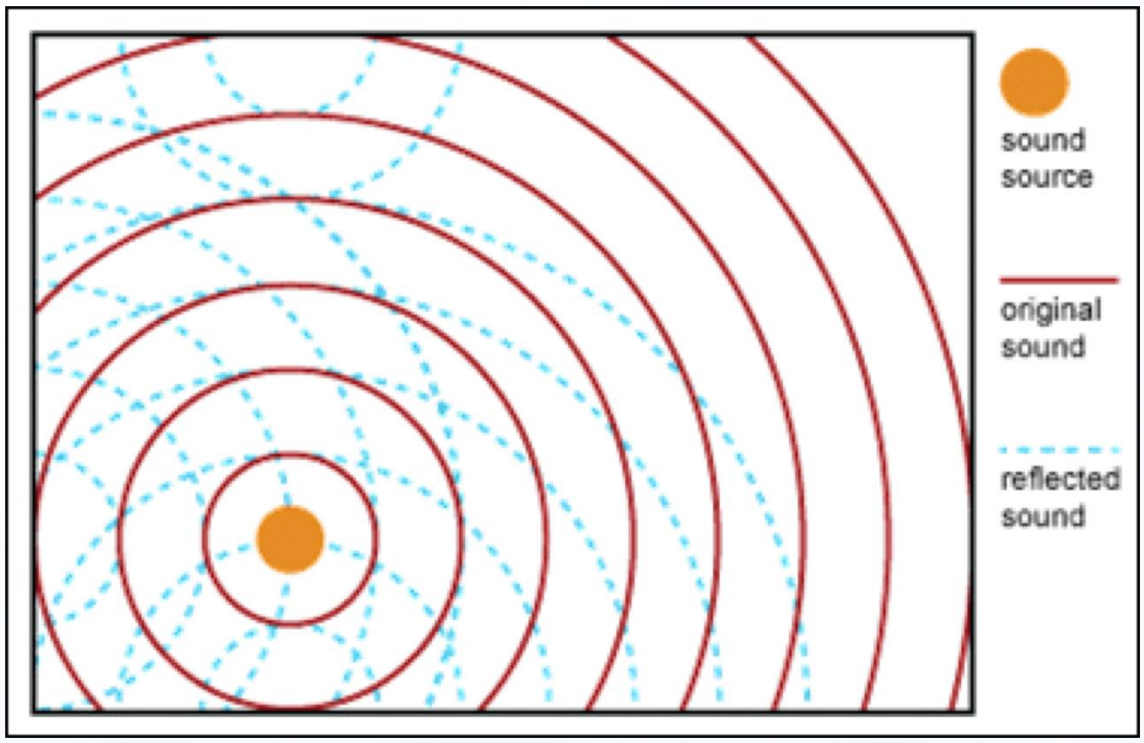
= 343 m/s

- **How Does Sound Behave?**

  Sound waves will undergo certain behaviors when it encounters the end of an obstacle. These behaviors include reflection off a boundary, diffraction around an obstacle, and even transmission through an obstacle making it a new medium.

Reflection occurs when sound waves bounces of walls and flat hard surfaces. The more contrasting properties between the boundary surface and that of the sound wave the more reflection occurs. Inversely the more similarities between the surface and sound the lesser reflection occur. The sound wave is then absorbed by the surface and converted as heat. Natural phenomenon caused by reflection is those that we perceived as echo and reverberation.

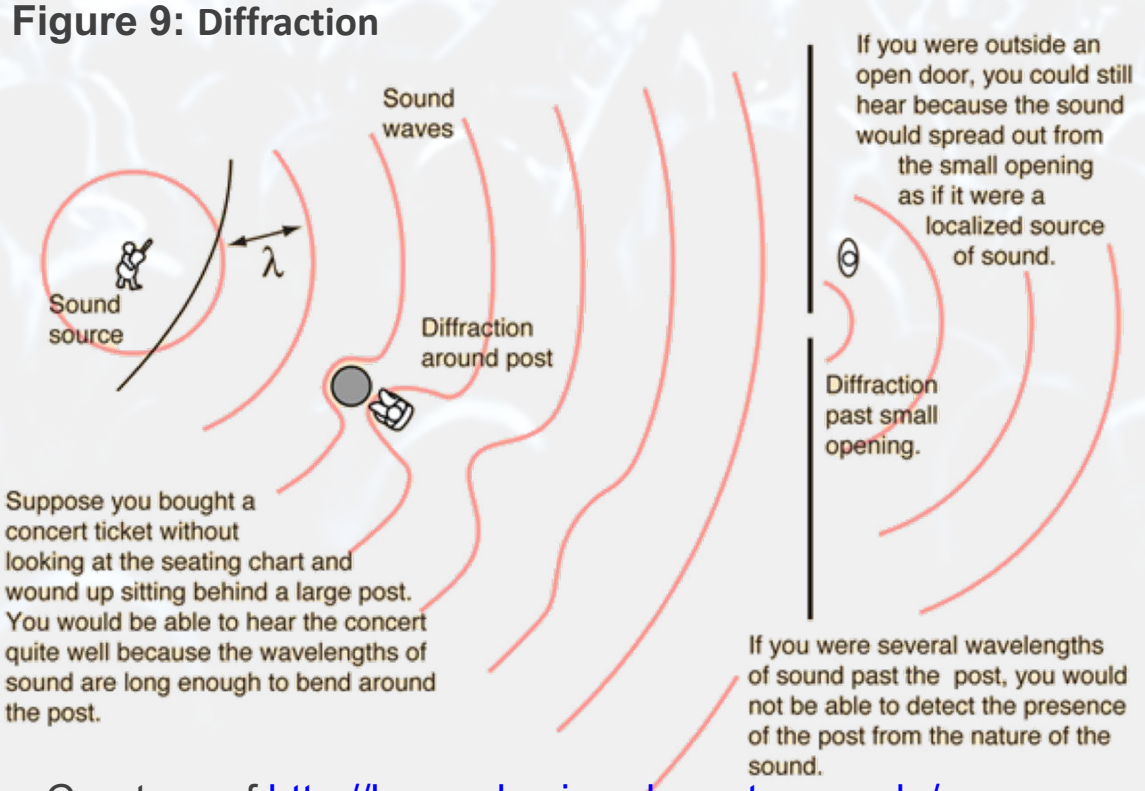
Figure 8: Reflection



Courtesy of https://www.osha.gov

Diffraction is the change in direction of waves as they pass through an opening or around an obstacle along its path. Just like water or air sound waves can reconstruct or assumes a container's shape. Diffraction occurs easier at lower wavelengths.

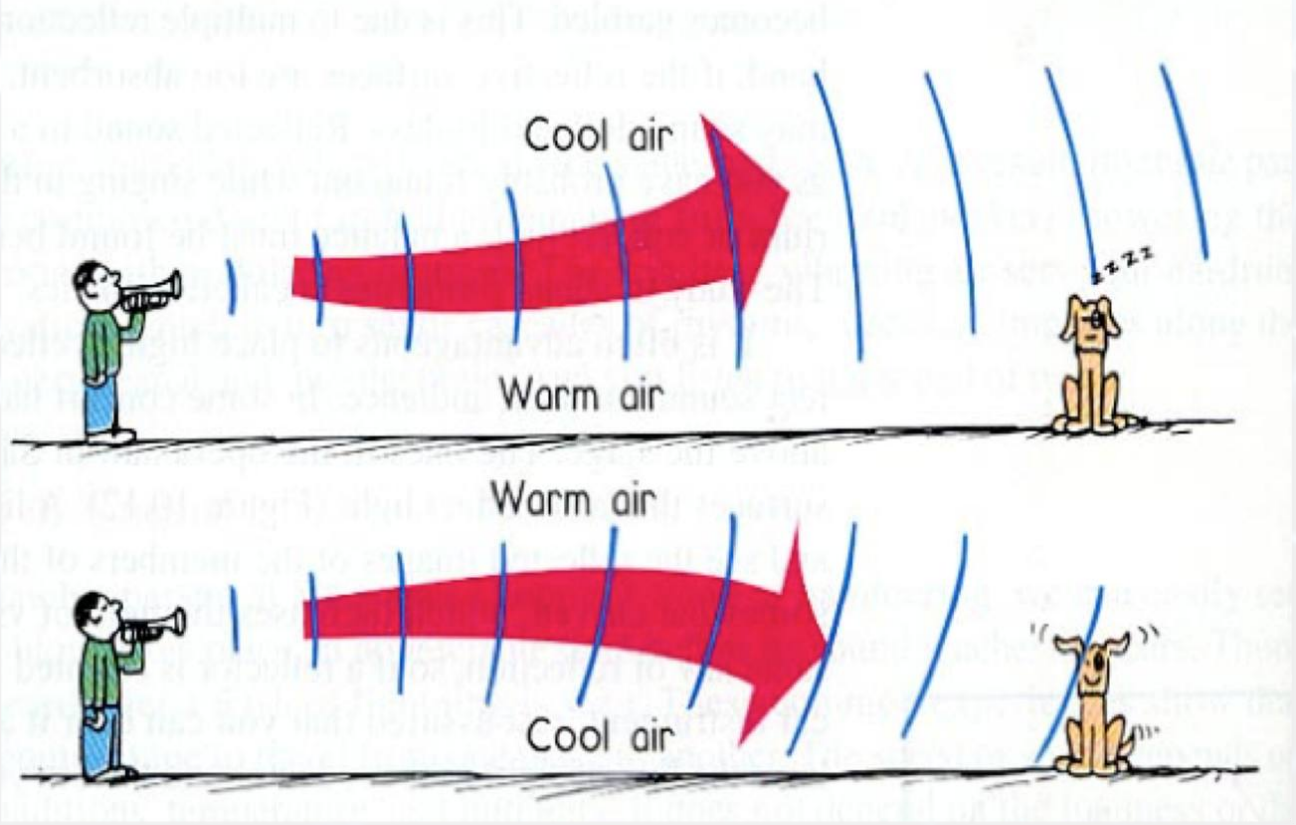
Figure 9: Diffraction



Courtesy of http://hyperphysics.phy-astr.gsu.edu/

Refraction is a change in the direction of waves as they pass from one medium to another. It is the bending of the path of the waves is accompanied by a change in speed and wavelength of the waves. One example is when the path of sound passes over a river. Water is known to cool down the air above it, so in effect the sound that is passing above it will be slower than the sound passing higher above with higher temperature. Since cold air under warm air tends to go downward the sound along with it refracts downward as well.

Figure 10: Refraction



Courtesy of https://www.thermaxxjackets.com

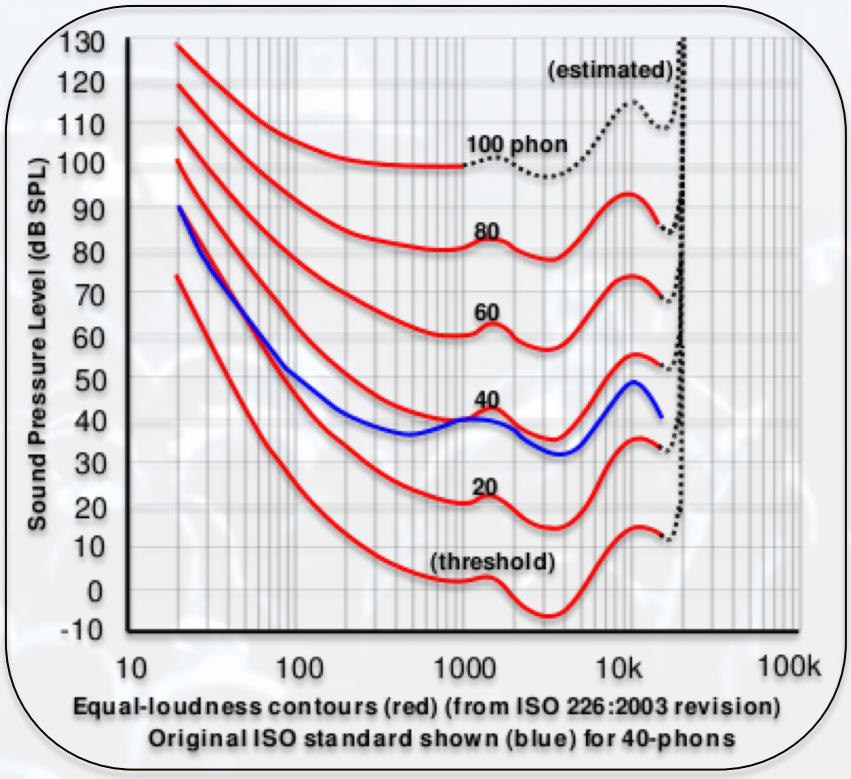
Loudness Level and the “Threshold of Pain”

My recommended Loudness setting is based on this chart

- **First Row 95 dB** (or plus 20 dB above ng ambient noise)
- **Back Row 85 dB** (or plus 10 dB above ng ambient noise)
- The most ideal for me a consistent **90 dB SPL map**

Sound Source	Decibels
Jet aircraft, 50m away	140
Threshold of pain	130
Threshold of discomfort	120
Chainsaw, 1m distance	110
Disco, 1m from speaker	100
Diesel truck, 10m away	90
Curbside of busy road, 5m away	80
Vacuum cleaner, 1m away	70
Conversational speech, 1m away	60
Average home’s room tones	50
Quiet library	40
Quiet bedroom at night	30
Background in TV studio	20
Rustling leaves in the distance	10
Hearing threshold	0

Source: Tontechnil-Rechner



“Equal-loudness contours”

Based on the Fletcher Munzon study the closest we could hear all frequencies evenly is anywhere between **75 to 85 dB SPL**

It is for these reasons that
“I almost always never go over and above 100 dB SPL”

Before we go on to the actual step by step procedure we must come to the understanding that….

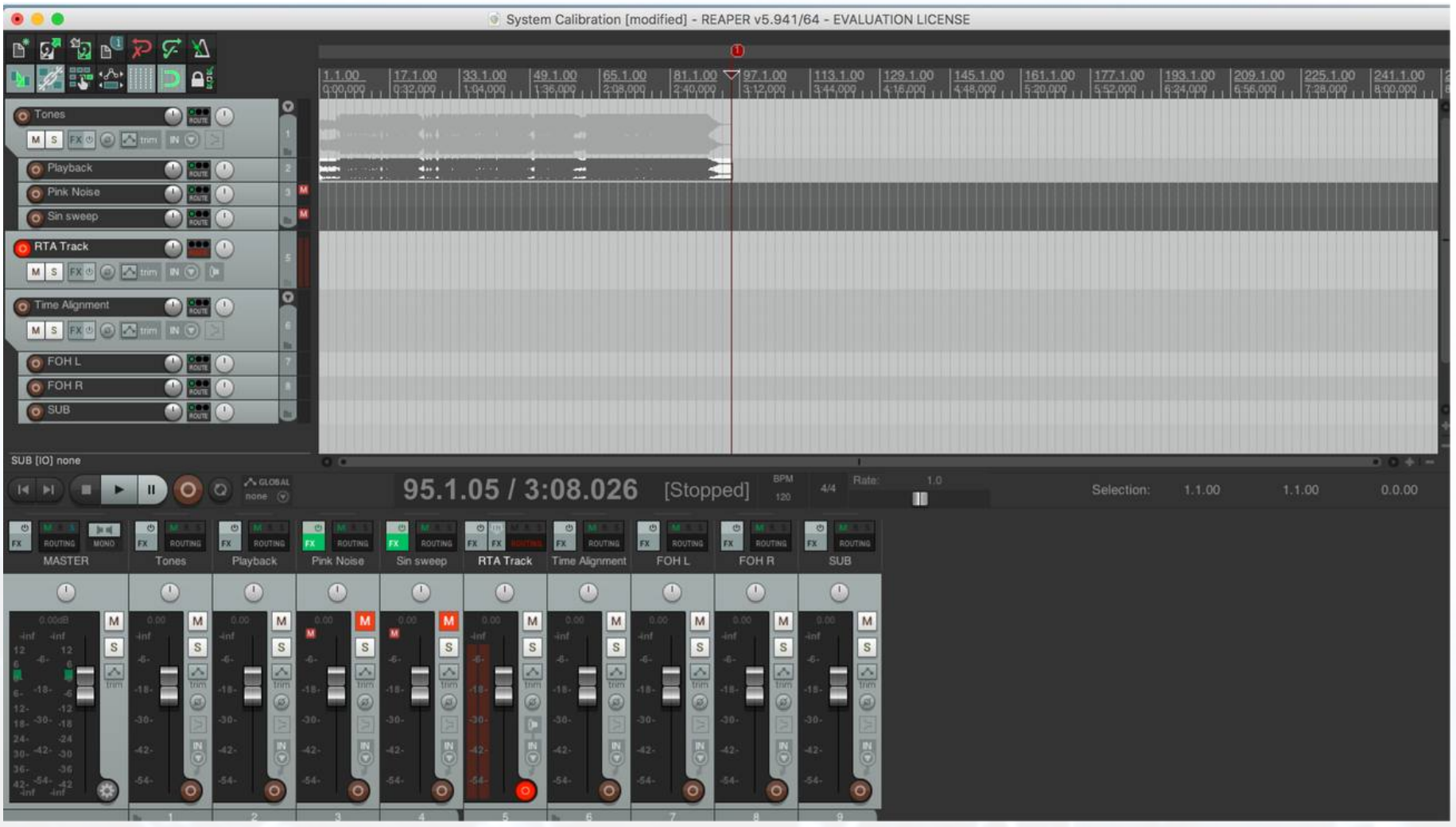
1. **There is a limit to our hearing capacity** before we reach the feeling of discomfort and God forbid, hearing damage.
2. **There's also a certain sweet spot** level where we can actually hear things correctly
3. **Sound behavior is dependent on many physical variables** within any given space.
4. **Loudspeaker response quality is only as good as the room** or environment where it is situated.
5. **We really could not rely on mere tasteful listening** to objectively quantify and judge sound when it comes to calibration.

All those theoretical stuff you read up until this point are the basis for the step by step process that I developed. Basically my 4 Step process to Calibrating a Sound System are:

1. **Optimized loudspeaker placement -** to ensure dispersion energy is not wasted
2. **Proper exposure levels -** by measuring the ambient noise to determine how loud the operating levels should or will be
3. **Time aligned loudspeakers -** to ensure clarity and intelligibility
4. **Spectral balance compensation -** by analyzing and comparing source audio material against the acoustical energy being produce by the system

Creating a Template

Your template should look like this once its finished.



Click the Reaper logo to download and install the demo version!
The DEMO version is ALL YOU NEED!

REAPER
Digital Audio Workstation

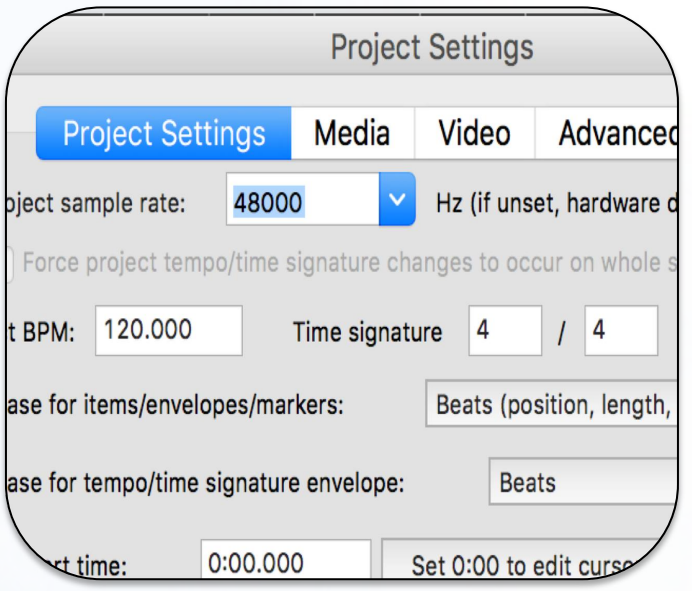
https://www.reaper.fm/download.php

Creating a Template

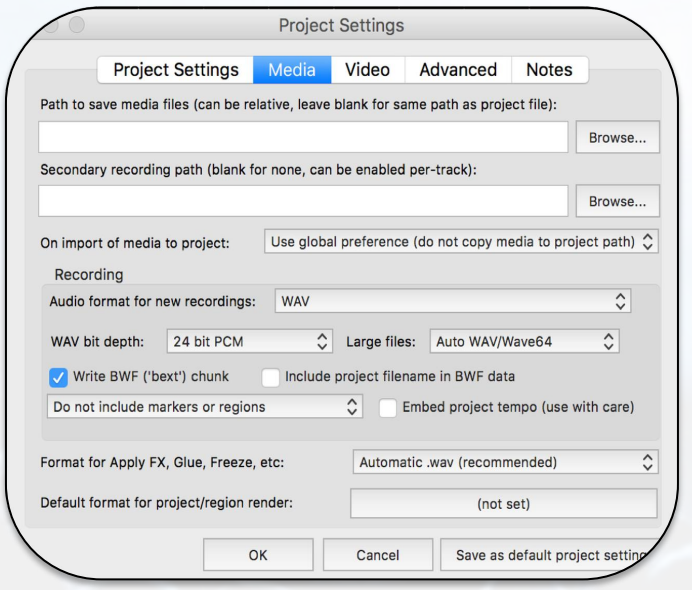
1) Create new project



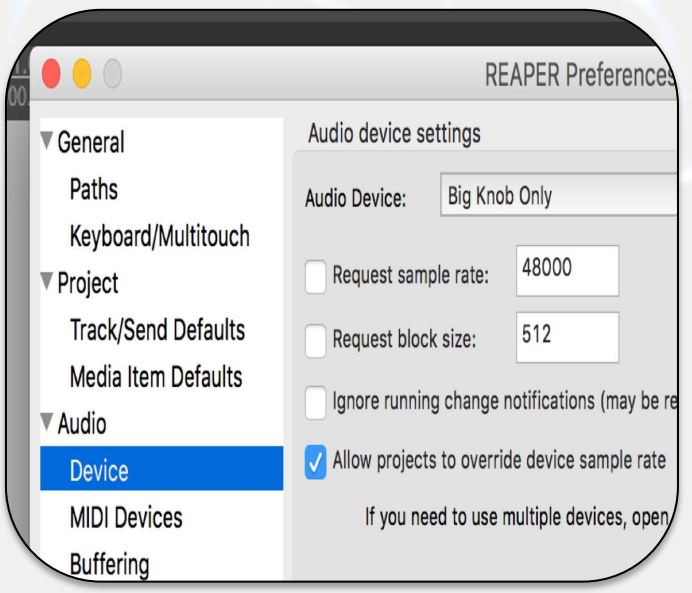
✓ Set the sample rate to at least 44.1khz



✓ Set the bit depth to 24 bit

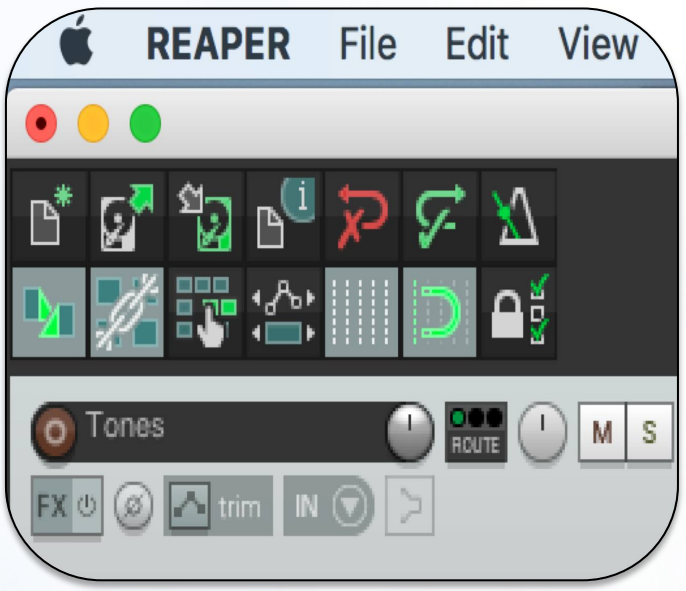


✓ Configure the audio device and choose ASIO drivers for windows (for mac just select the audio driver)



Creating a Template

2) Create a “Tones” folder track



✓ Create a “Tones” track



✓ Create a “Playback” track
✓ Drag and drop it to the Tones track



✓ Insert any audio file, preferably any music that has a decent mix of frequency content.

Creating a Template

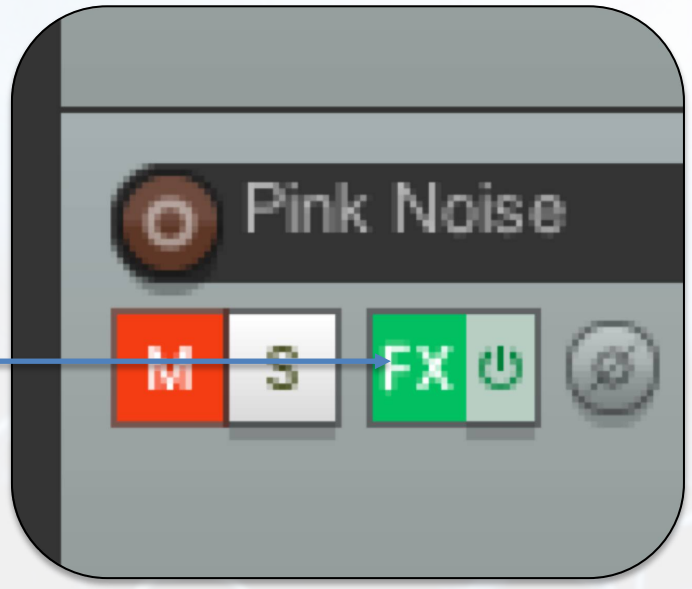
2) Create a “Tones” folder track

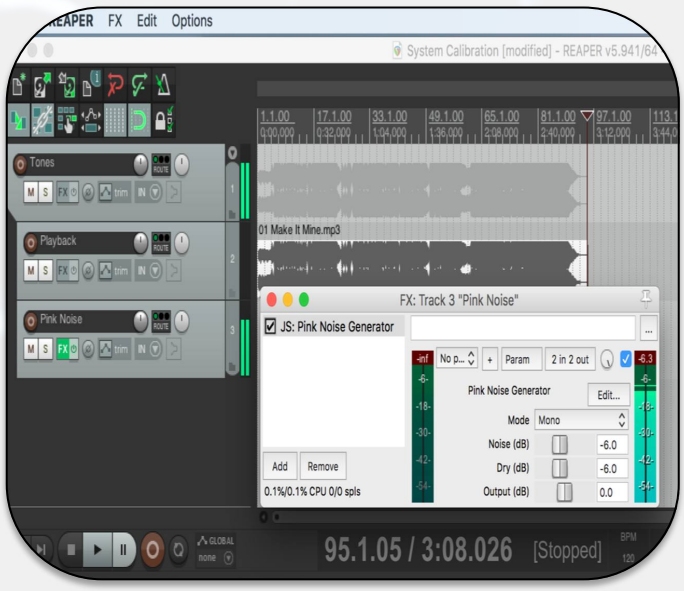
- ✓ Create a “Pink Noise” track
- ✓ Drop it to the Tones track
- ✓ Mute the track

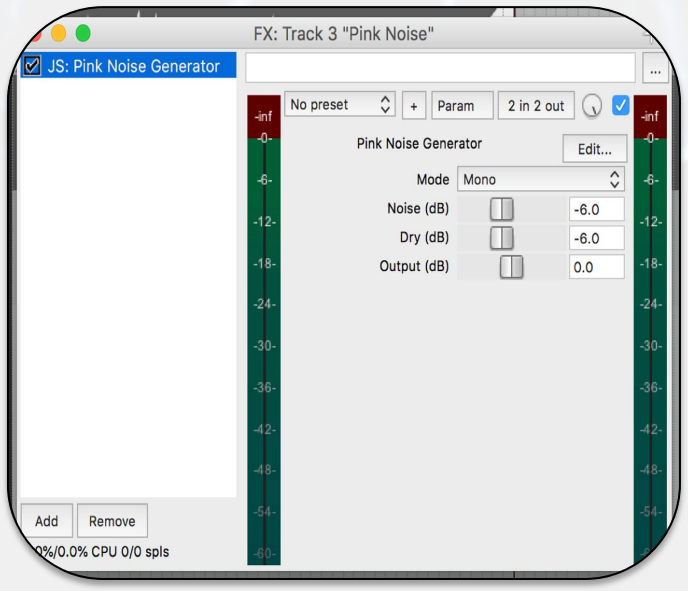




- ✓ Insert a pink noise plugin by clicking the FX button





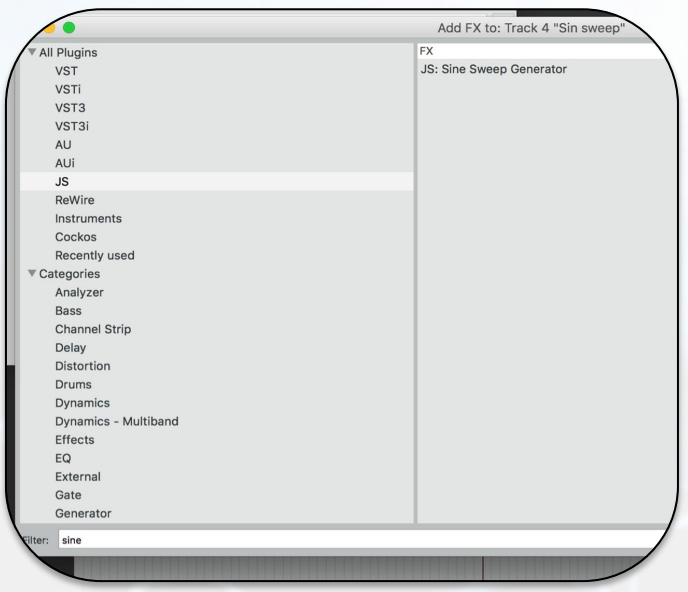


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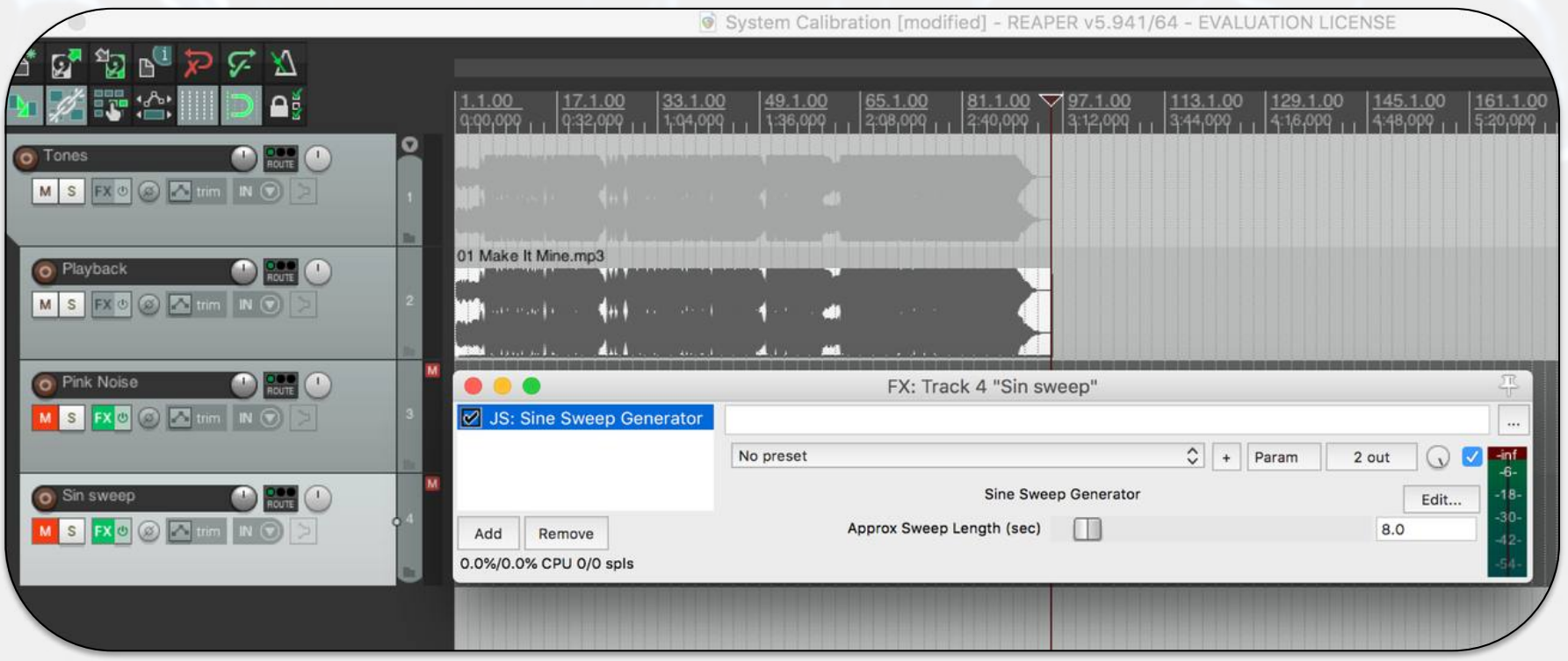
2) Create a “Tones” folder track

- ✓ Create a “Sine Sweep” track
- ✓ Drop it to the Tones track
- ✓ Mute the track





- ✓ Insert a Sine Sweep plugin



Creating a Template

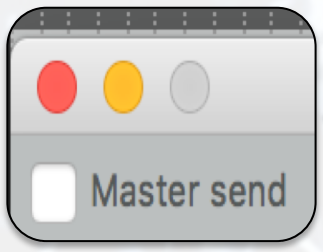
2) Create a “RTA” track



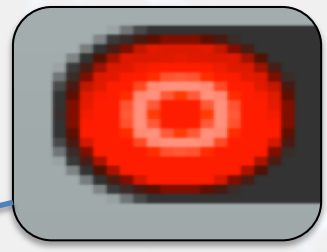
- ✓ Create a new track for the RTA mic







- ✓ Click the routing button
- ✓ Uncheck the Master Send tick Box





- ✓ Click the record enable
- ✓ Click the speaker icon



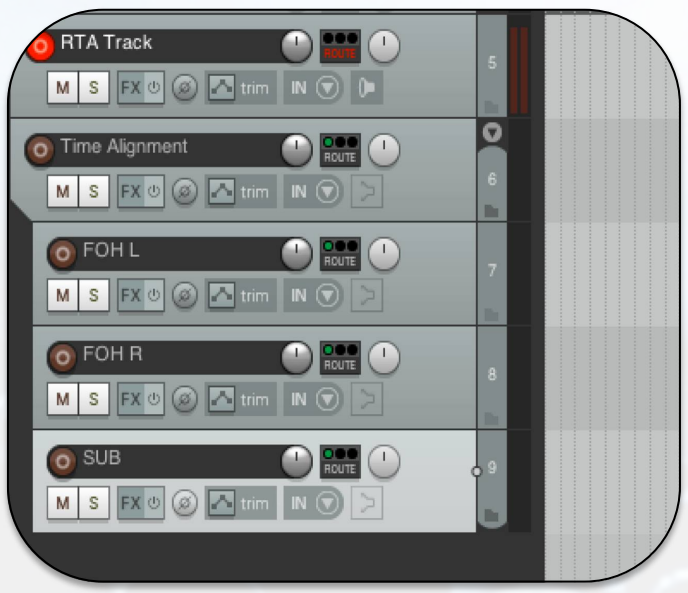
Creating a Template

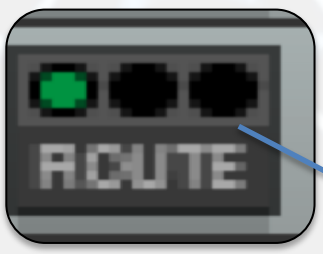
3) Create a “Time Alignment ” folder track

- ✓ Create new track for FOH L
- ✓ Create new track for FOH R
- ✓ Create new track for the subwoofer

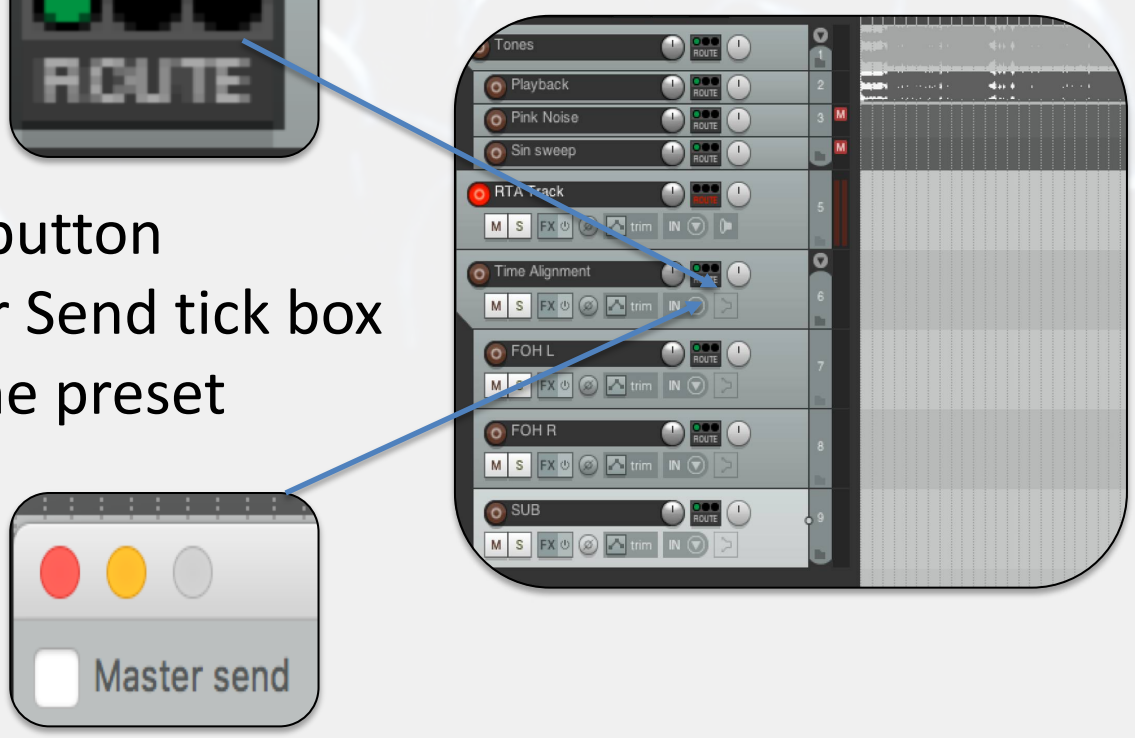


- ✓ Drop all tracks to the Time alignment folder track





- ✓ Click the routing button
- ✓ Untick the Master Send tick box
- ✓ Save and name the preset

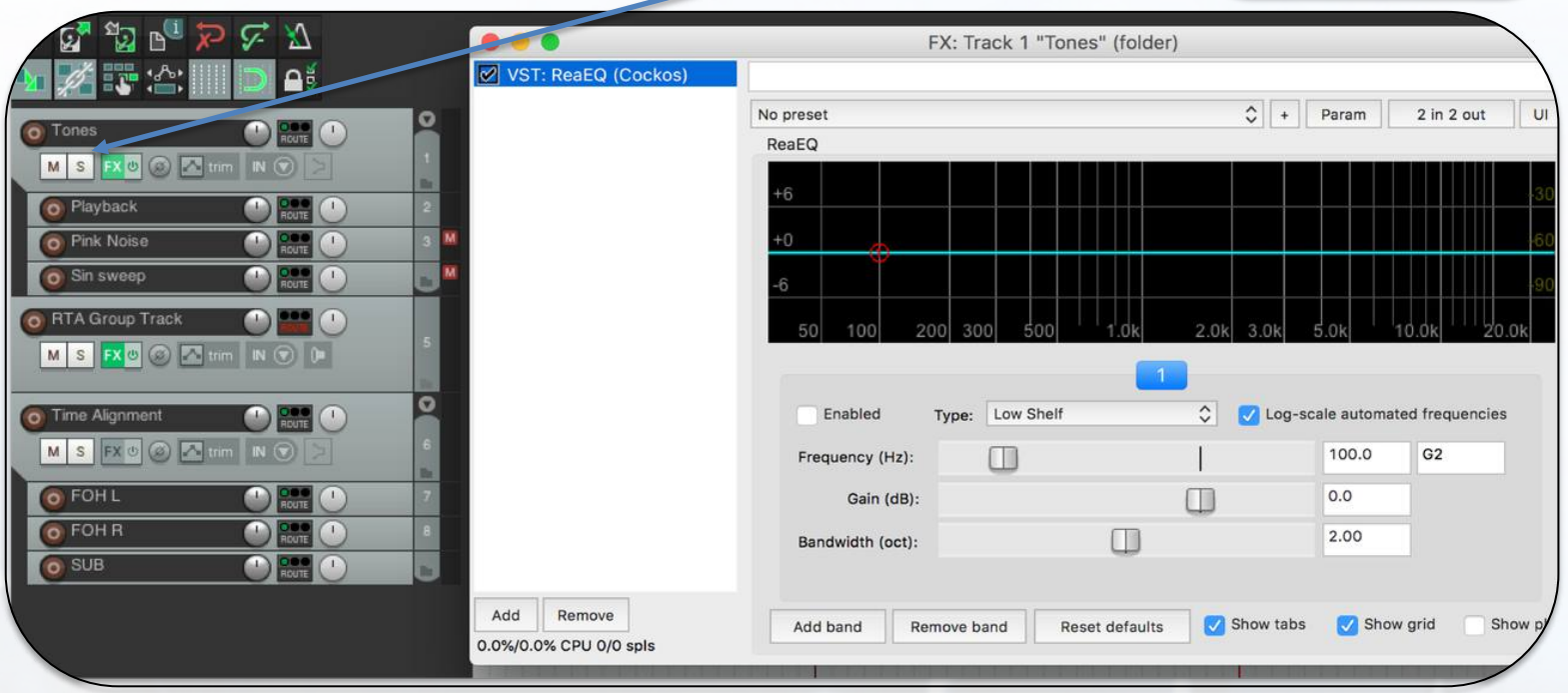


Creating a Template

4) Insert FFT Analysis Plugins

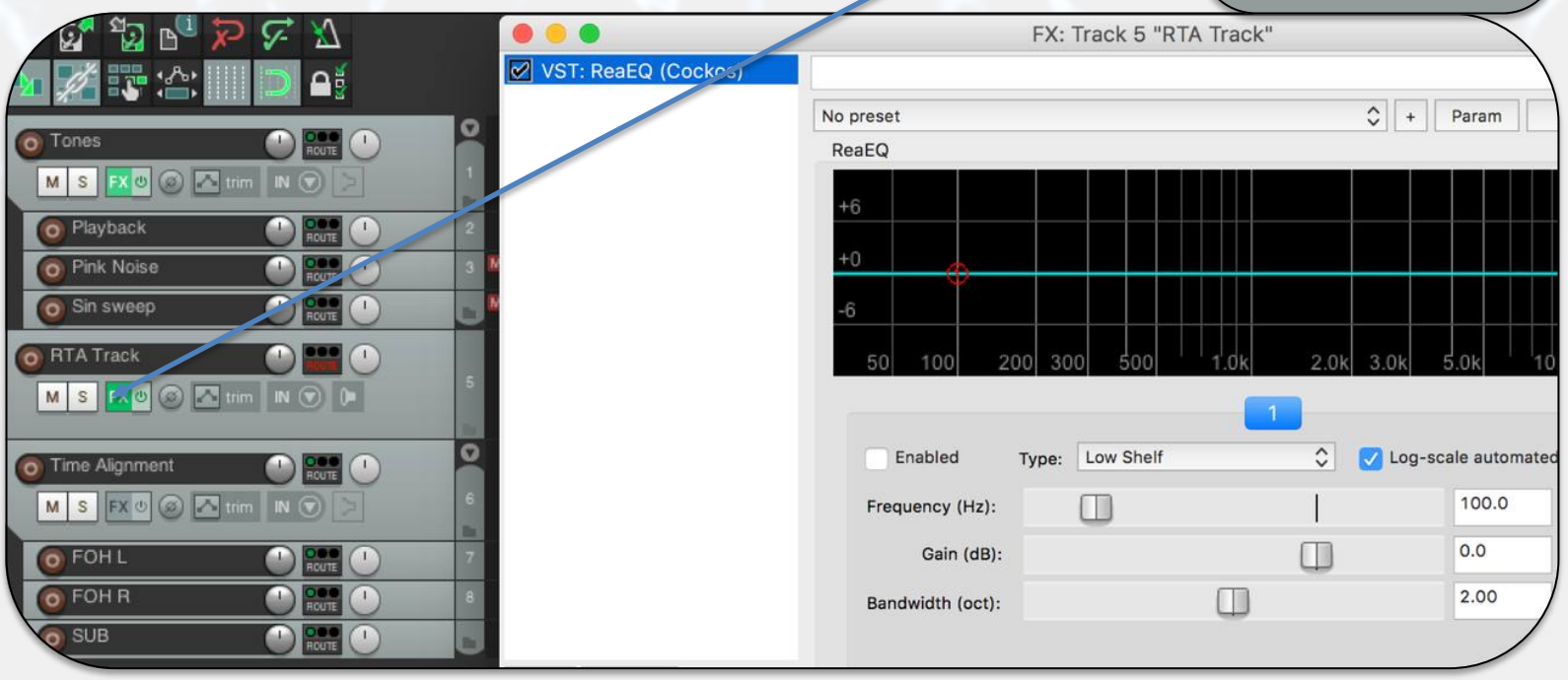


- ✓ Click The Tones FX button
- ✓ Insert ReaEQ plugin





- ✓ Click the RTA FX button
- ✓ Insert ReaEQ plugin



Hardware and Software Requirements

There is a reason why this book is sub titled **“Without Breaking Your Bank Account”** simply this method save you money from paying expensive appointments for consultant or buying expensive software that does the same things. There are just few hardware that will have to invest on if you don’t have them yet. Most of these devices can be acquired inexpensively as opposed to buying **“commercially bundled solutions”** which can really be expensive. In this method you will only be need the following.

RTA Mic



SPL Meter



Audio Interface.



Reaper D.A.W.



Hardware and Software Requirements

RTA Mic



This mic will serve as the ear that listens to the frequencies more accurately than our ears ever can. It is impervious to psycho acoustics (which gives us the tendency to be selective in what we hear).
I find that the make and model of the mic is insignificant as long as it is of decent brand. RTA measurement mics are all designed to just pick up a flat response on an omni-directional pattern. Frankly most of them that market this type of mic with their own brand are most probably built from the same manufacturing plant anyway.

SPL Meter



Basically this device quantifies loudness level, again it’s a machine that is impervious to psycho acoustics. Having this tool keeps us from second guessing exposure levels.

Hardware and Software Requirements

Audio Interface



An audio interface is a device that is intended to send and receive audio signals that will be used to test and measure electro acoustical data. If you are using a digital console however, you won't be needing this device since most digital consoles these days are designed to act as an audio interface for D.A.W. connectivity.



Digital Audio Workstation



The software analysis tool that we will be using is a Digital Audio Workstation. My personal choice is Reaper. You can use any D.A.W. as long as they can offer the same routing function and offers similar plugins. The Demo version of Reaper alone is good to go!

Hardware and Software Requirements

Diagram using a Digital Mixer

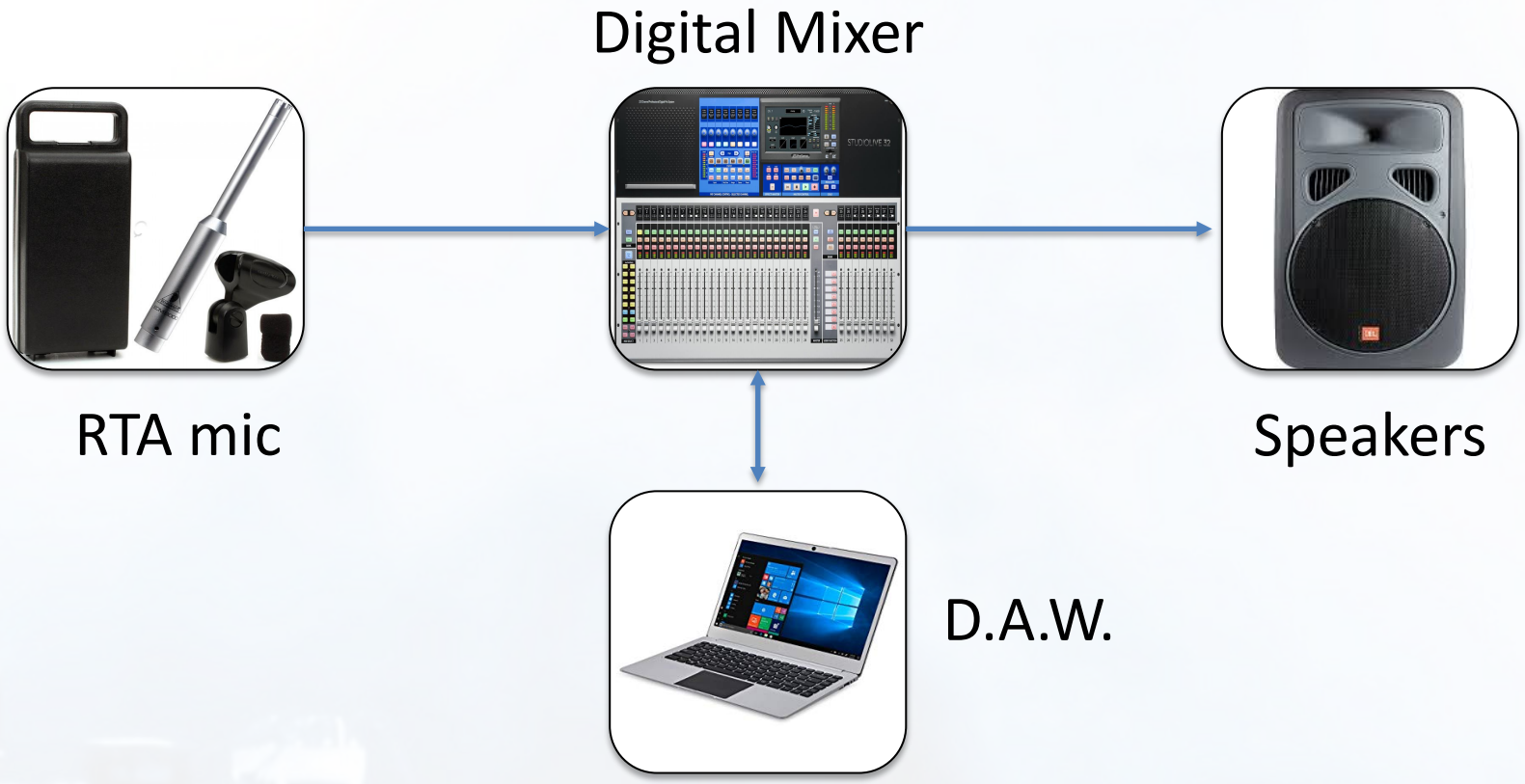
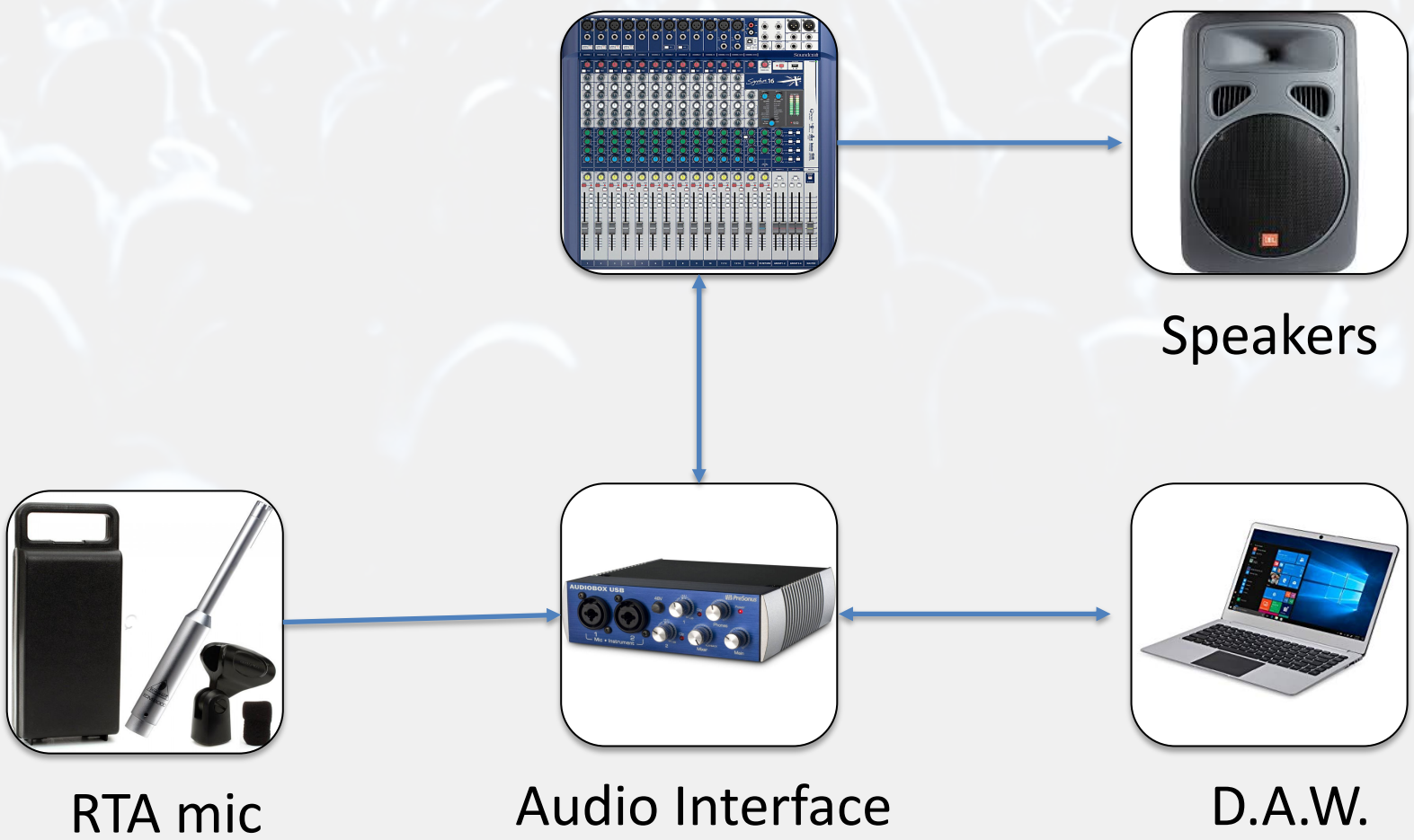
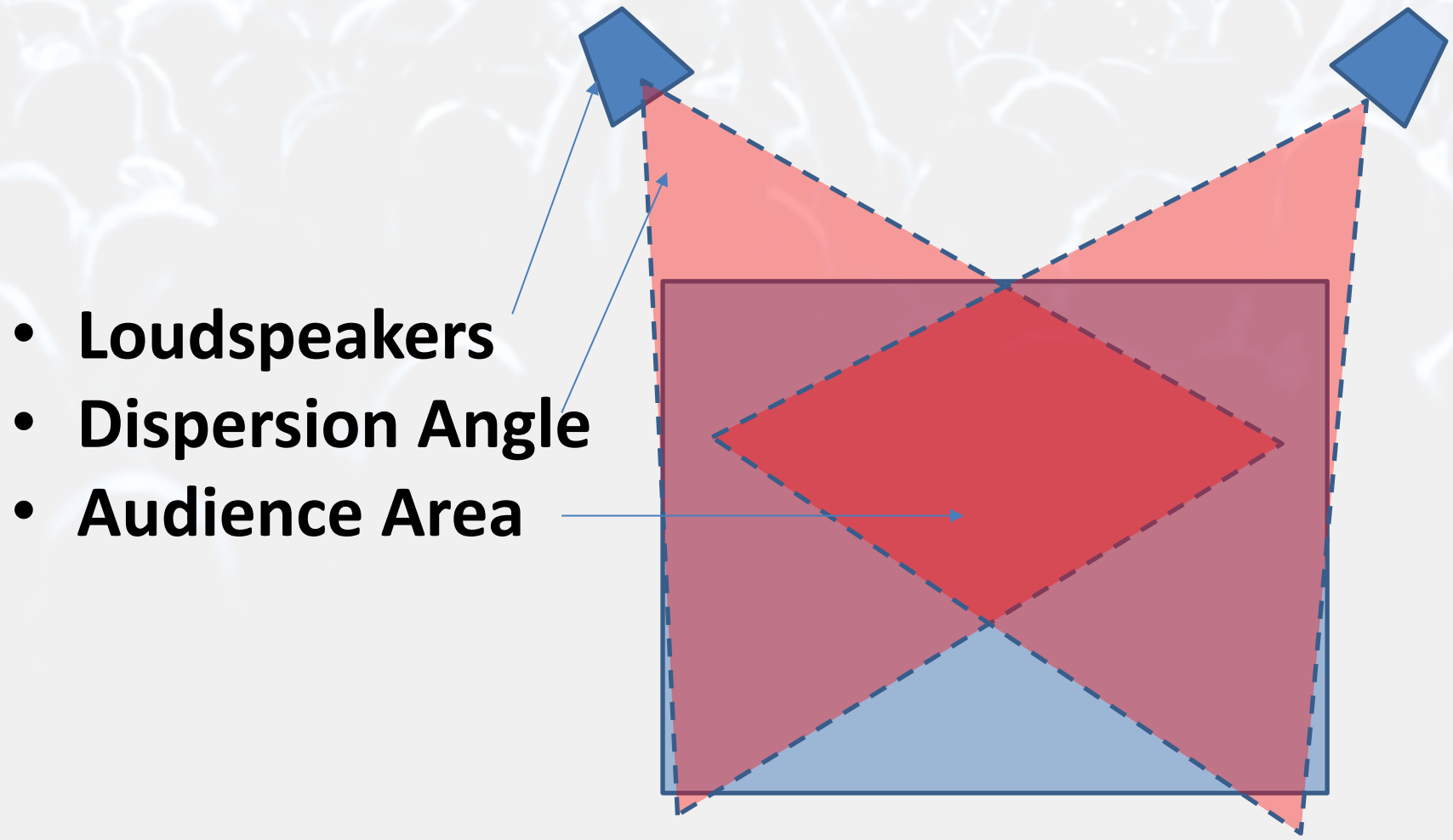


Diagram using an Analog Mixer



Proper Speakers Placement

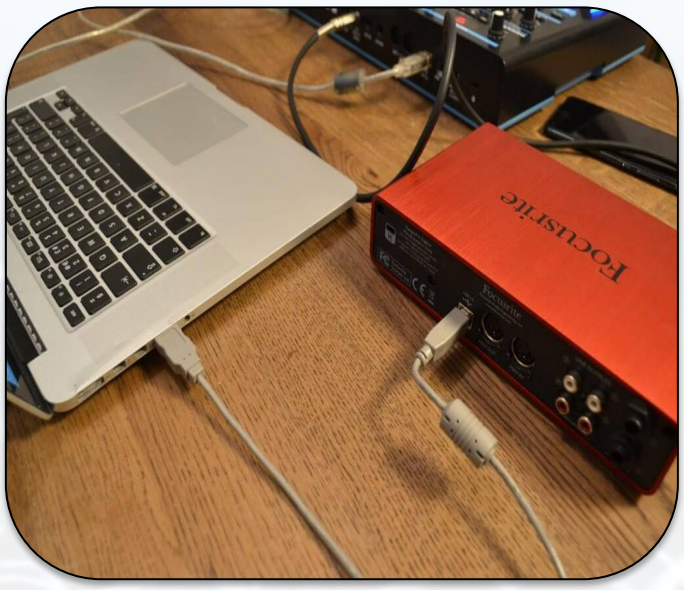
- ✓ Check the coverage or dispersion angle of your loudspeakers provided in the product documentation
- ✓ Position the loudspeakers in such a way that the dispersion angle is optimized to cover the audience area
- ✓ The goal is to get an even coverage of the whole audience area. Most live sound mixing is summed in the middle of the field. Stereo mixing is not really a concern. Whether you are running a bi-mono or a mono daisy chain signal lines, essentially you are still mixing in mono. Unless your goal is for the right side to hear only what's on the right and the left side likewise only hears the left side then go ahead mix in stereo.



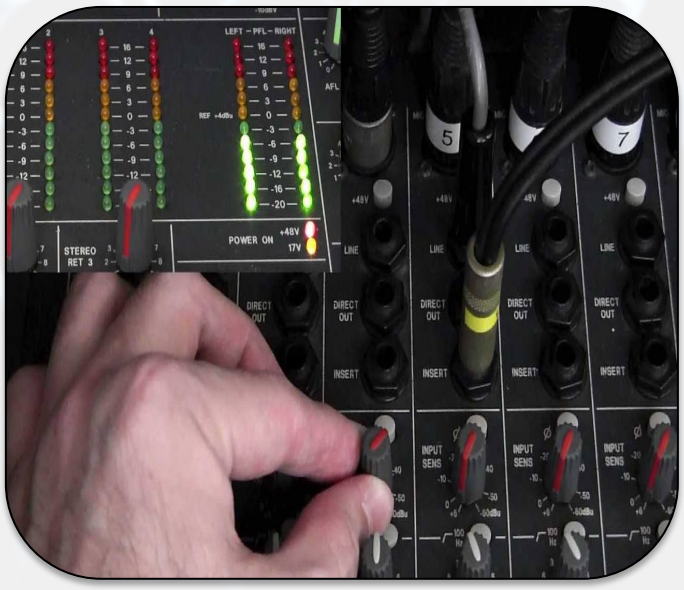
Proper Loudness Setting



- ✓ Measure and take note of the ambient noise
- ✓ Set the SPL meter to C-weighing mode
- ✓ Set it to Slow Response

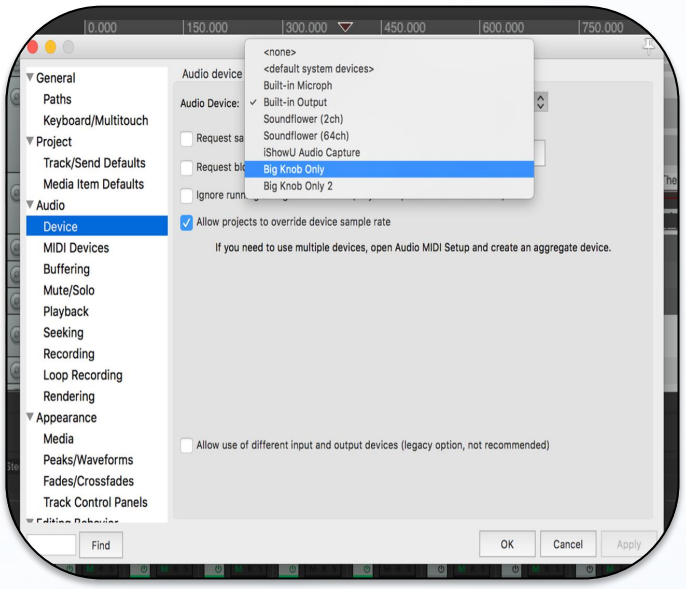


- ✓ Setup the D.A.W. (Computer, audio interface, Reaper)
- ✓ Patch the output of the D.A.W. into a channel on your mixer

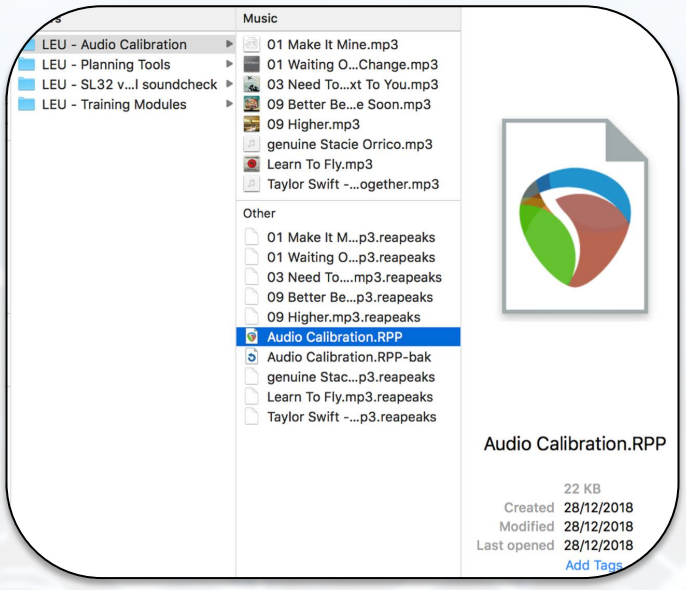


- ✓ Set all faders and knobs to unity gain
- ✓ Set the D.A.W channel fader to 0 dB

Proper Loudness Setting



- ✓ Launch Reaper and set the mixer/interface to use the ASIO driver
- ✓ Assign the inputs and output accordingly



- ✓ Open File “Audio System Calibration.RPP”



- ✓ Unmute track labeled pink noise to inject audio into the system

Proper Loudness Setting



- ✓ Set the meter to C- weighing mode
- ✓ Set the response time to slow
- ✓ Measure the SPL level



- ✓ Adjust the output faders until you hit the target SPL
- ✓ Mark or save the fader position

Ninja tip! you can also find out your system's headroom capacity by pushing the faders higher as possible without overdriving the speakers. Lets say your SPL reading is at 90dB with your output faders at 0 dB, if you raise the fader + 10 dB, your meter would read around 100 dB. This means your system has twice the loudness capacity. Just pull the faders back to your target SPL. Now you have confidence that your system can still go up 10 dB more if in case you need to later on the show. If however you are unable to reach the target SPL level, it only means that your system doesn't have enough power to handle the venue.

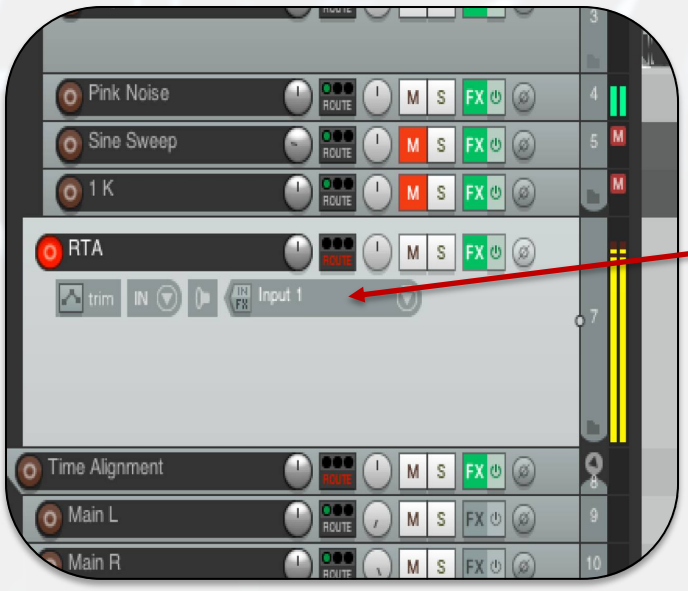
Loudspeaker Time Alignment



- ✓ Set the RTA mic at the “reference position ” around ear height (or average ear height)



- ✓ Connect the RTA mic to a channel on your digital mixer or audio interface
- ✓ Make sure that there is no processing applied to that channel
- ✓ Do not assign the RTA to any output (or simply mute it)
- ✓ Engage the phantom power

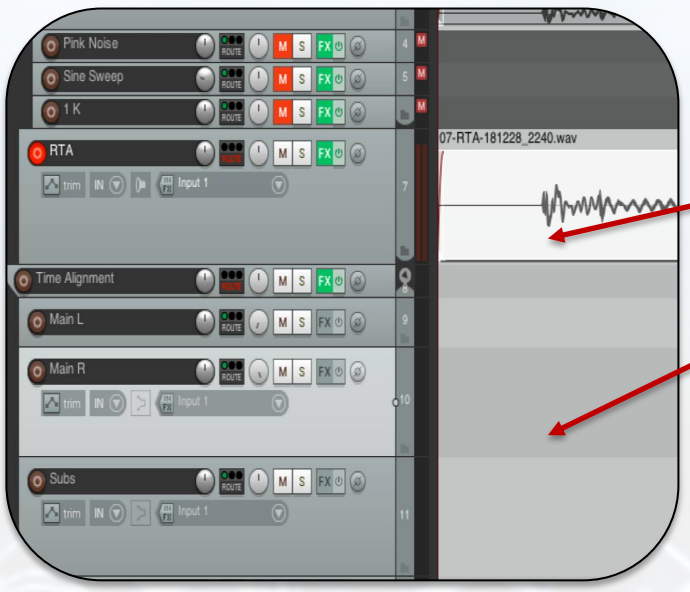


- ✓ Set the RTA input in Reaper that corresponds to the input channel where the RTA mic is plugged into

Loudspeaker Time Alignment



- ✓ Set the playback curser of Reaper at the start of the timeline
- ✓ Solo the farthest speaker (FOH) from the reference point
- ✓ Click the record button
- ✓ After 10 seconds click the stop button

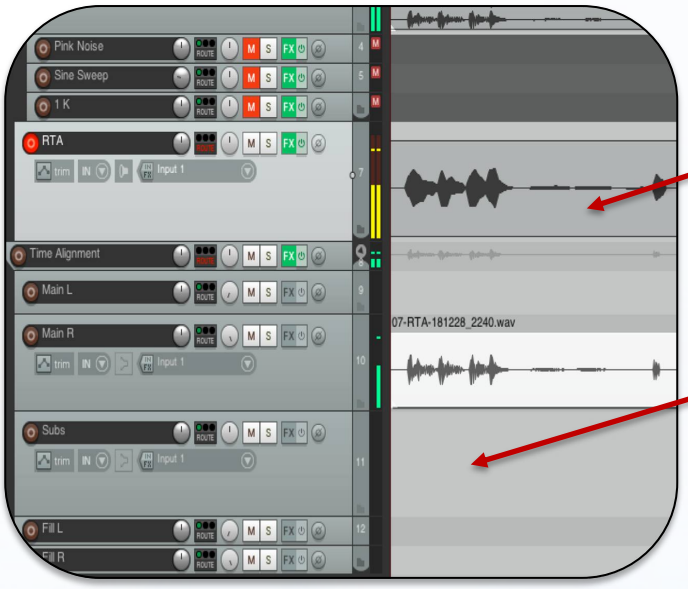


- ✓ Drag and drop the recorded RTA track to the “FOH” track
- ✓ Make sure that clip is dropped at the beginning of the timeline

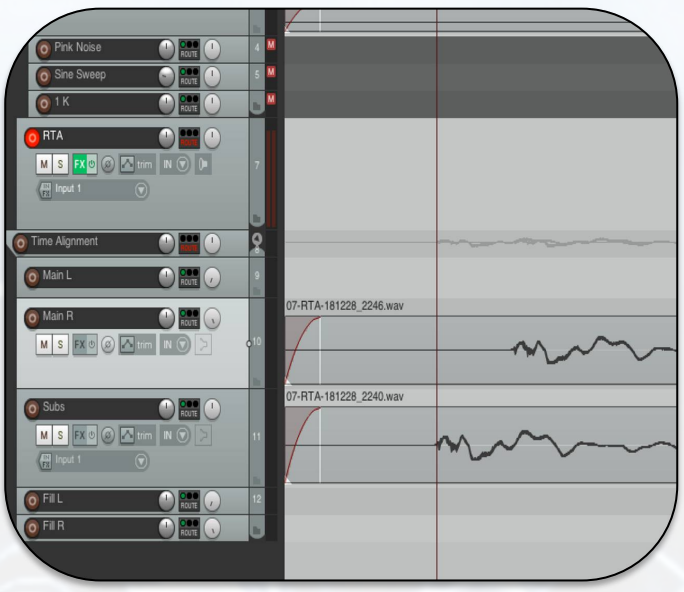


- ✓ Set the playback curser back to the start of the timeline
- ✓ Solo the speaker that is closer (Sub or back fill) to the reference point
- ✓ Click ang record button
- ✓ After 10 seconds click the stop button

Loudspeaker Time Alignment



- ✓ Drag and drop the recorded RTA track to the “SUB Woofer” track
- ✓ Make sure to drop it at the beginning of the timeline

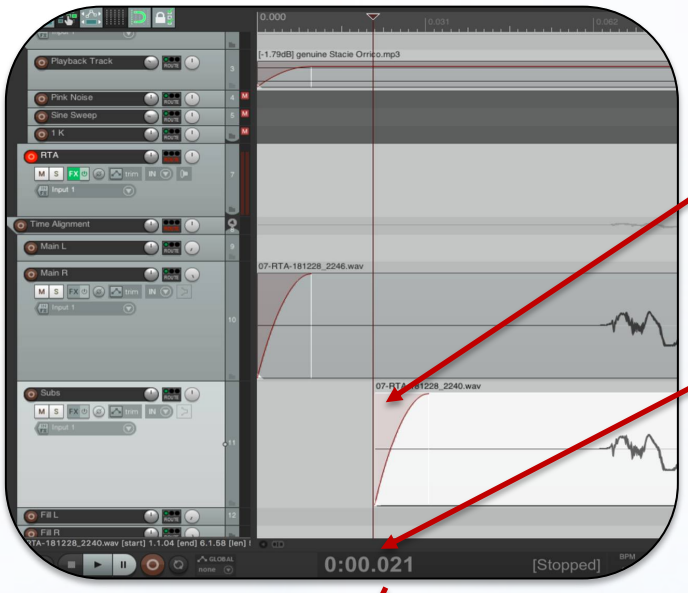


- ✓ Compare the two waveforms by enlarging the resolution



- ✓ If there is a difference between the start of the record times of the two clips, move the clip that was recorded at an earlier part of the timeline

Loudspeaker Time Alignment





- ✓ Place the **curser at the start time** of the waveform that was nudged
- ✓ Take note of the **“Time Difference”**
- ✓ Convert seconds into ms
  - Example: .032 seconds to 32 ms (.032X1000)
- ✓ Apply a time delay to the output of the speaker that needs to be delayed
- ✓ Use the time value that you got from editing the waveform position. (**“Time Difference”)**

Ninja tip! Repeat the whole process depending on the number of speakers that needs to be aligned. The reference should always be the latest speaker that is recorded on the timeline, in most cases that would be the farthest speaker from the reference point.

Spectral Balance



✓ Unmute the Pink Noise Track

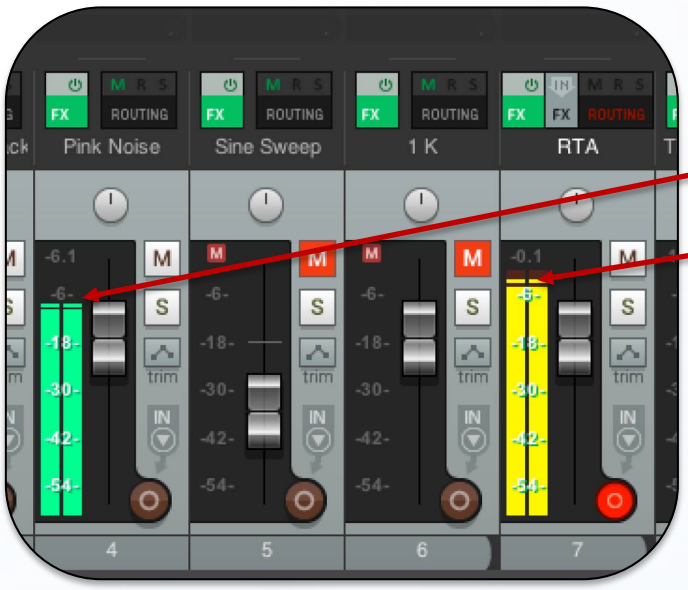


✓ Mute the FOH fader



✓ Feed the audio to the Sub-Woofer output

Spectral Balance

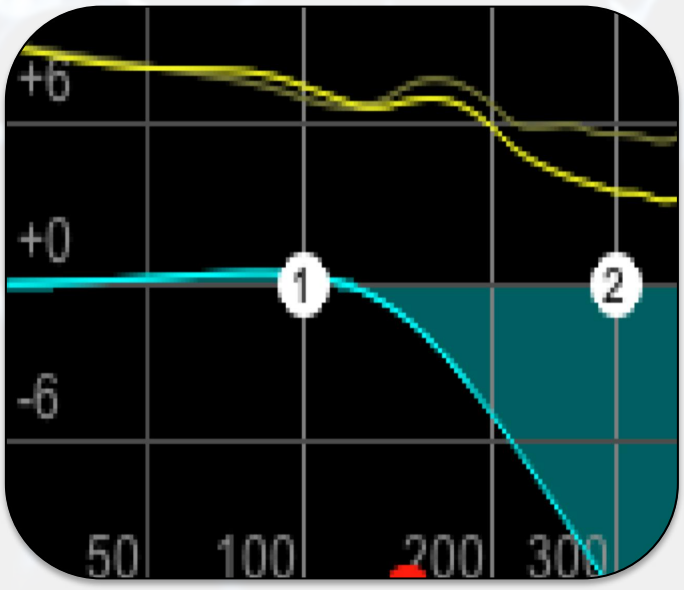


- ✓ Match the input and output levels of the signal by adjusting the RTA input





- ✓ Click the FX plugin of the RTA track to view the FFT analyzer



- ✓ Apply a “Low Pass Filter” to the output of the Sub-Woofer according to the recommended response based on the user manual

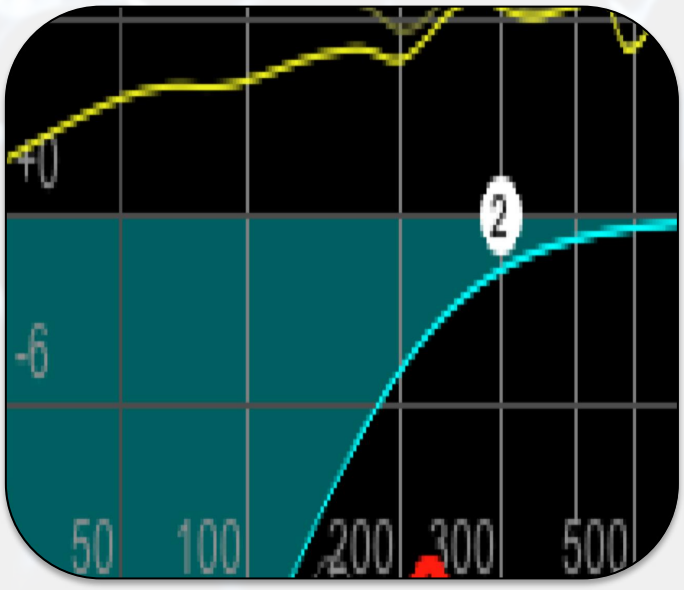
Spectral Balance



- ✓ Mute the Sub-Woofer
- ✓ Unmute the F.O.H



- ✓ Click the FX plugin of the RTA track view the FFT analyzer

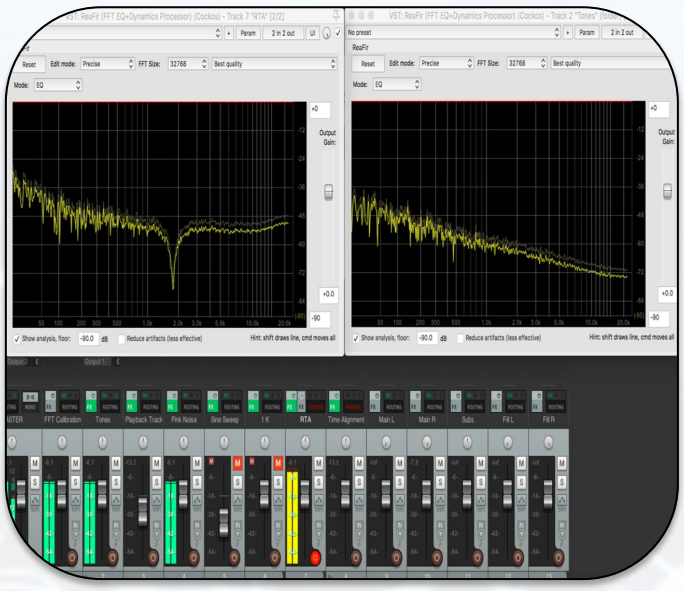


- ✓ Apply a “High Pass Filter” to the FOH output
- ✓ Set the cut-off according to the cut-off you applied to the Sub-woofer (Crossover Point)

Spectral Balance



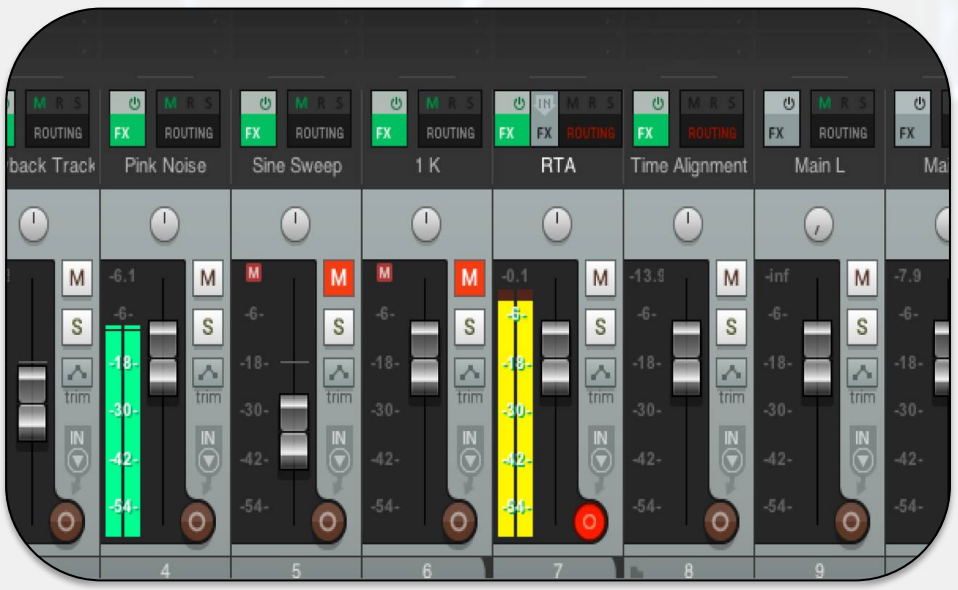
- ✓ Unmute both the F.O.H. at Sub-Woofer faders



- ✓ Click the FX plugin of the **RTA** to view its FFT analyzer
- ✓ Click the FX plugin of the **Tones** to view its FFT analyzer

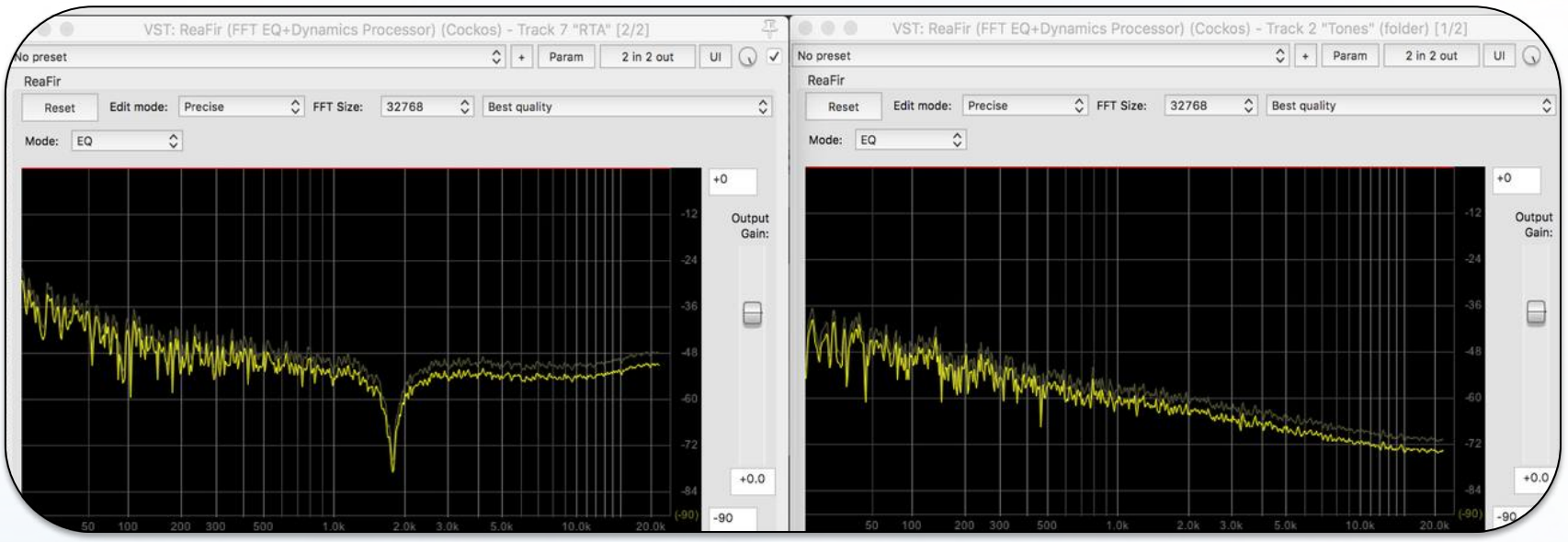
- ✓ Gain match the input and output once again by adjusting the RTA input gain





Spectral Balance

- ✓ Compare the FFT readings of the Pink Noise and of the RTA

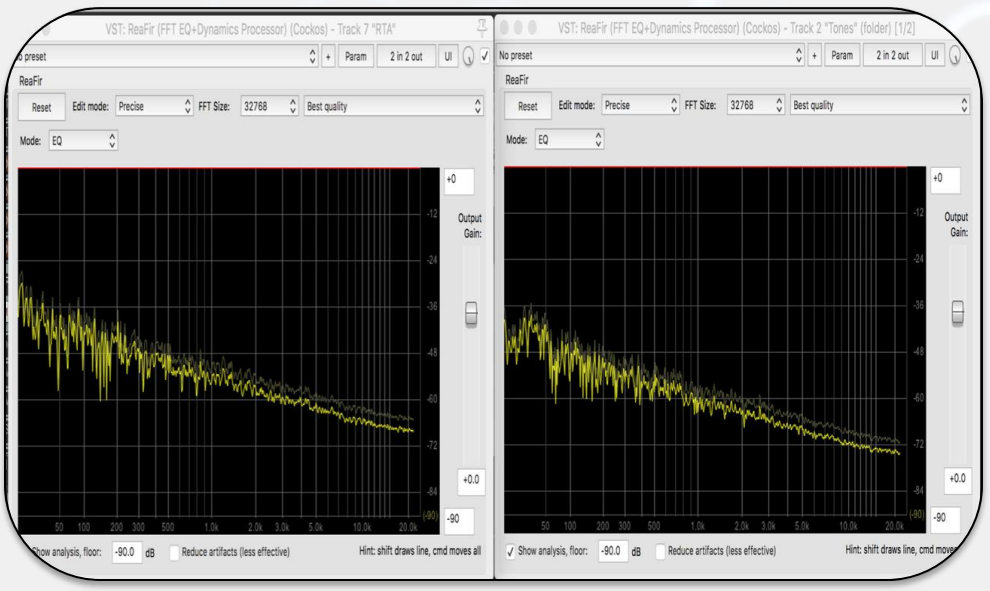




- ✓ Adjust the fader blend between the F.O.H and the Sub-Woofer until the RTA and Pink Noise reading matches as closes as possible

- ✓ Adjust the E.Q. of F.O.H. (or even the Sub-Woofer if need be) to fine tune the response until the RTA and Pink Noise readings are almost identical.





Spectral Balance

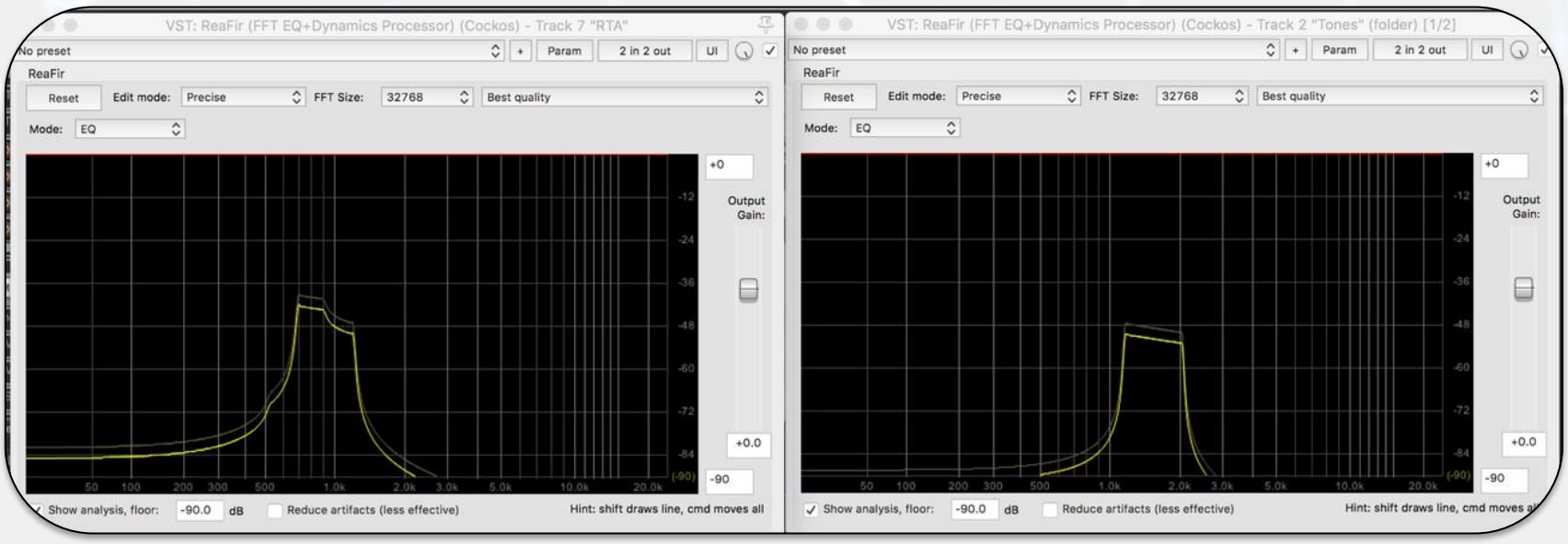


- ✓ Mute the Pink Noise track
- ✓ Unmute the Sine Sweep track



- ✓ Compare the RTA and Sine Sweep reading
- ✓ Fine tune the E.Q. more if need be

Ninja tip! 2 to 3 dB is kind of negligible difference you may leave it at this point. When you see more than 6 dB difference however you might want fine tune the E.Q. more.





My name is **Bone Pagligaran,** the dude behind audioninjatv.com. I have been involved in the local audio and music industry in the last 25 years as a musician, sound engineer, music producer, system integrator, and audio consultant. In the past couple of years I have been more involved in 1 on 1 and group coaching in audio production. If you are interested in a live exclusive training please contact us @ support@audioninjatv.com

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